The empirical recurrence for OEIS A276303, proved by transfer matrix

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Abstract

OEIS A276303 counts arrays in which no entry may equal the entries at a short list of fixed offsets from it, the arrays being counted up to renaming the letters. The entry carries an empirical recurrence of order 30 contributed by R. H. Hardin. It is true, and it is decidable rather than empirical. The condition tests only whether two cells are equal, so it survives renaming; the count of classes is recovered from counts over fixed alphabets by inverting a triangular system of falling factorials, and each of those is a walk count on a window of 1 consecutive columns, $S = 797$ states in all.

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1 The conjecture

OEIS A276303 is "Number of 6Xn 0..2 arrays with no element equal to any value at offset (-2,-1) (-1,1) or (0,-1) and new values introduced in order 0..2.". It has offset 1 and begins

$$122, 250, 3437, 34865, 442001, 4988145, 59728757, 693817033, \ \ldots$$

The entry states, as a conjecture contributed by its author and never marked settled:

> Empirical: a(n) = 3*a(n-1) +78*a(n-2) +250*a(n-3) +558*a(n-4) -937*a(n-5) -4256*a(n-6) -7991*a(n-7) -3472*a(n-8) +31381*a(n-9) -11144*a(n-10) +42683*a(n-11) -198936*a(n-12) +231086*a(n-13) -131301*a(n-14) +70502*a(n-15) -27827*a(n-16) +159651*a(n-17) -217780*a(n-18) +120250*a(n-19) -55920*a(n-20) -17125*a(n-21) +26805*a(n-22) -25487*a(n-23) +30226*a(n-24) -15183*a(n-25) +5503*a(n-26) -1930*a(n-27) +371*a(n-28) -65*a(n-29) +8*a(n-30) for n>32

As of the "Last modified" line on the live entry (Aug 28 2016 08:30:47, revision 4) this is still recorded as empirical, and nothing on the entry records it as proved.

2 What is being counted

Entries lie in $\{0, 1, \ldots, 2\}$ and the condition is that

$$x(i,j) \ \neq \ x(i + d_1,\, j + d_2) \qquad \text{for each offset } (d_1, d_2) \in \{(-2,-1), (-1,1), (0,-1)\}$$

whenever the second cell lies inside the array. The entry writes the array with n as the number of COLUMNS, so the array grows one COLUMN at a time and an offset is read with its two coordinates exchanged. The closing clause "new values introduced in order 0..2" says the array is the canonical representative of its equality pattern.

Lemma 1. *The number counted does not depend on the order in which the array is read.*

Proof. The condition tests only whether two cells carry equal values, so it is unchanged by permuting the alphabet, and the admissible arrays fall into classes under renaming. Fix any reading order of the cells. Each class contains exactly one array whose distinct values, in order of first appearance in that reading, are $0, 1, 2, \ldots$: take any member and rename its values by their rank of first appearance. So the count is the number of CLASSES, whatever reading order is meant. $\square$

3 From classes to fixed alphabets

Write N_j for the number of classes on exactly j letters and L_i for the number of admissible arrays over an alphabet of i letters. An array over i letters is a class together with an injection of its letters into the alphabet, so

$$L_i \;=\; \sum_j N_j\, i(i-1)\cdots(i-j+1),$$

a triangular system with unit diagonal. By Lemma 1 the entry counts $N_1 + \cdots + N_3$, so $a = \sum_i w_i L_i$ with $w = \mathbf{1}^\top F^{-1}$, F the matrix of falling factorials.

4 Each L_i is a walk count

Equality is symmetric, so an offset and its negative forbid exactly the same pairs; replacing each offset by whichever of the two points BACKWARDS along the direction in which the array grows makes every pair tested once, from the later of its two cells. After that replacement no offset reaches more than 1 columns back, so the window of the last 1 columns is a state and one step appends a column. A column not yet written is carried as a symbol, so the walk starts at the empty array and a walk of t steps is an array of t columns. Assembling $i = 1, \ldots, 3$ as a disjoint union with ι carrying the weights cleared of denominators and $\tau = \mathbf{1}$,

$$a(n) \;=\; \iota^\top M^{\,j}\tau, \qquad S = 797.$$

The walk index j and the entry's own n differ by a constant, read off by matching the published terms rather than assumed. In particular a is C-finite.

5 The proof

Lemma 2. *Let $q(t) = t^{30} - \sum_i c_i t^{30-i}$ be the characteristic polynomial of the conjectured recurrence. The residual $a(n) - \sum_i c_i a(n-i)$ equals $\iota^\top M^{\,j} q(M)\tau$ for the corresponding walk index j, so the recurrence holds from some point on if and only if $\iota^\top M^j q(M)\tau = 0$ for all large j, and it is enough to examine $j < S$.*

Proof. Factor M^j out of $M^{j+30} - \sum_i c_i M^{j+30-i}$. For the bound, M^S is an integer combination of $I, M, \ldots, M^{S-1}$ by Cayley–Hamilton, so the numbers $u_j = \iota^\top M^j q(M)\tau$ obey the monic recurrence given by the characteristic polynomial of M; S consecutive zeros force every later one, and the last nonzero u_j pins the threshold exactly. $\square$

Theorem 3.

$$a(n) \;=\; 3\,a(n-1) + 78\,a(n-2) + 250\,a(n-3) + 558\,a(n-4) - 937\,a(n-5) - 4256\,a(n-6) - 7991\,a(n-7) - 3472\,a(n-8)$$

for every $n > 32$.

Proof. The vector $w = q(M)\tau$ was formed by 30 matrix–vector products in exact integer arithmetic and $\iota^\top M^j w$ evaluated for $j = 0, 1, \dots,$ again exactly; those integers vanish from the index corresponding to $n = 32 + 1$ onwards and the run continued until the working vector was identically zero. The rational weights are carried as one common denominator, so no division is performed until the final term. $\square$

6 Verification

Four checks, each independent of the algebra above.

First, the model reproduces all 16 terms the entry publishes, exactly, in integer arithmetic.

Second, the count was recomputed for the smallest arrays straight from the definition: enumerate every array over $\{0, \dots, 2\}$, keep those whose values are introduced in the order $0, 1, 2, \dots$ reading row by row, and test each offset directly. Neither the window, nor the backward normalisation of the offsets, nor the falling-factorial inversion is involved, and the published terms came back.

Third, the conjectured recurrence was evaluated directly on the published terms, with no matrices involved, and holds wherever the proved range and the data overlap.

Fourth, the annihilation test of Lemma 2 was carried out in exact integer arithmetic on the whole vector rather than on a sampled prefix.

References

[1] The OEIS Foundation, *The On-Line Encyclopedia of Integer Sequences*, `https://oeis.org`, 2026. Sequence A276303.

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[3] R. A. Horn and C. R. Johnson, *Matrix Analysis*, 2nd ed., Cambridge University Press, 2013.